

OCCUPANCY MEASURES! PERFORMANCE- DIFFERENCE LEMMA! POLICY OPTIMIZATION!

Gergely Neu



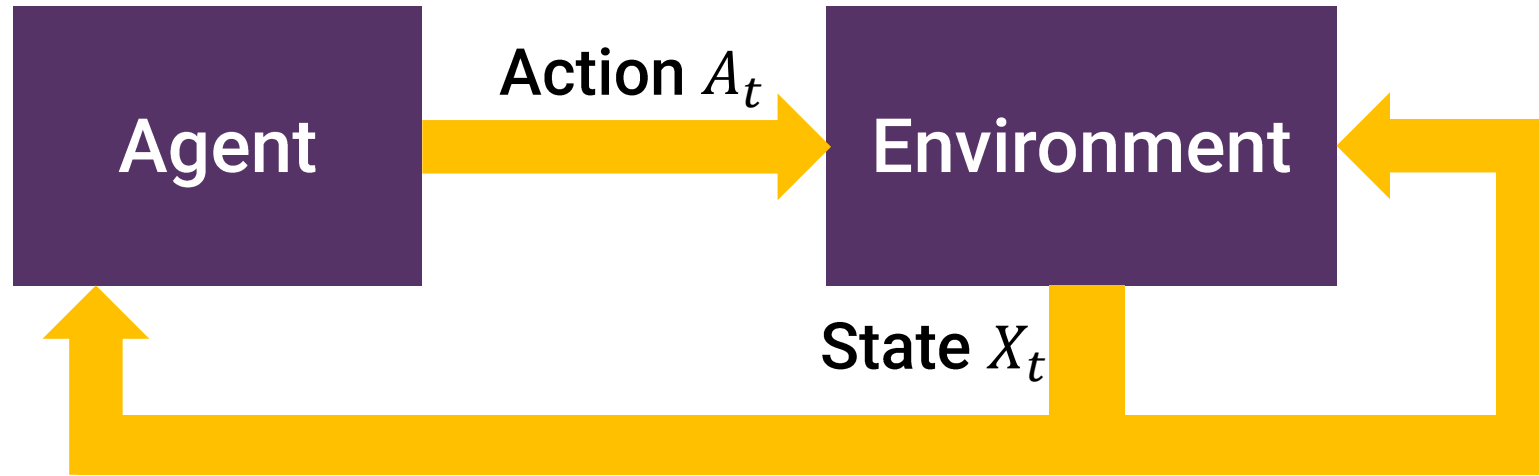
Universitat
Pompeu Fabra
Barcelona

OUTLINE

- Markov decision processes
- Occupancy measures
- The performance difference lemma
- Applications:
 - Policy improvement theorem
 - Policy gradient theorem
 - Soft policy improvement

MARKOV DECISION PROCESSES

MARKOV DECISION PROCESSES



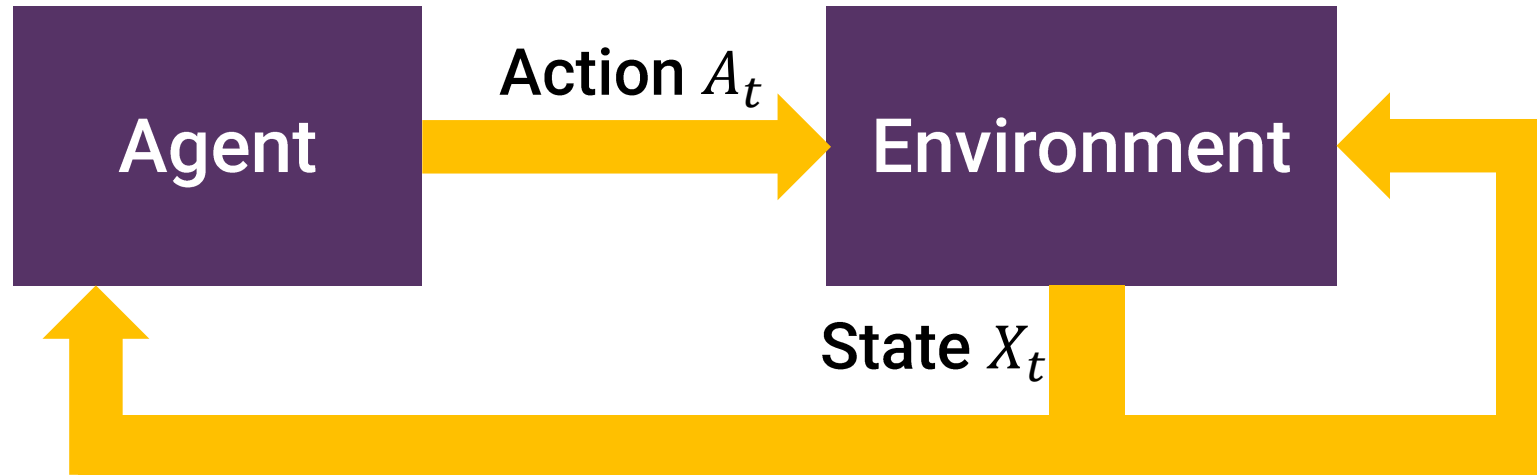
Agent:

- Observe state X_t , take action A_t
- Obtain reward $r(X_t, A_t)$

Environment:

- Generate next state $X_{t+1} \sim P(\cdot | X_t, A_t)$

MARKOV DECISION PROCESSES



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Goal:

maximize discounted return

$$\rho = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r(X_t, A_t) \right]$$

from initial state $X_0 \sim \nu_0$

$\gamma \in (0,1)$

BASIC MDP FACTS

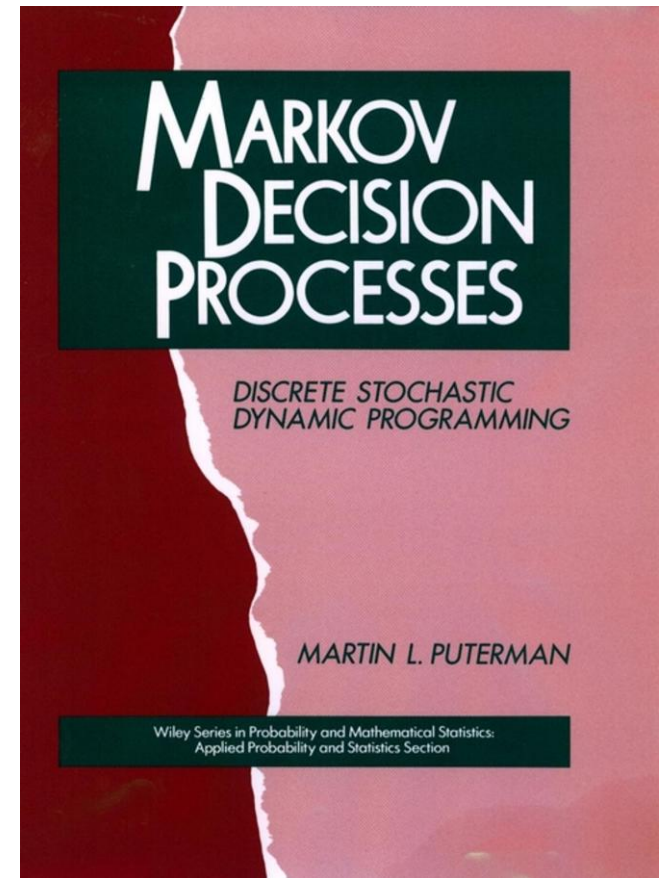
- Markov property: X_{t+1} only depends on (X_t, A_t)
- Stationarity: $P(\cdot | X_t, A_t)$ doesn't depend on t



enough to consider **stationary policies**

$$\pi(a|x) = \mathbb{P}[X_t = a | X_t = x]$$

- Many other beautiful properties:
 - There is a deterministic optimal policy
 - Simultaneous optimality regardless of v_0
 - ...



SOLVING MDPs: THE CANONICAL WAY

The Bellman optimality equations

$$Q^*(x, a) = r(x, a) + \gamma \mathbb{E}[\max_{a'} Q^*(x', a') | x, a]$$

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immediate reward

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Optimal policy can be extracted as:

$$\pi(a|x) = \begin{cases} 1 & \text{if } a = \arg \max_{a'} Q^*(x, a') \\ 0 & \text{otherwise} \end{cases}$$

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Richard Bellman (1954):
Solution can be found via
“**Dynamic Programming**”

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Foundation for “value-based RL methods”:

- Stochastic approximation: TD, Q-learning,...
- Approximate DP methods: LSTD, LSPI, DQN,...

OUR GOAL TODAY

study an alternative theory that can serve as foundation
to policy-based RL methods

OCCUPANCY MEASURES

A LINEAR REFORMULATION

Observe: the discounted return of policy π is

$$\rho^\pi = \mathbf{E}_\pi[\sum_{t=0}^{\infty} \gamma^t r(X_t, A_t)]$$

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A LINEAR REFORMULATION

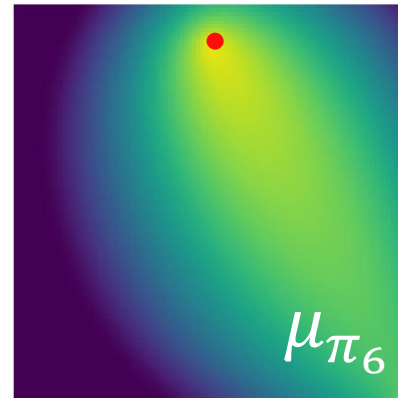
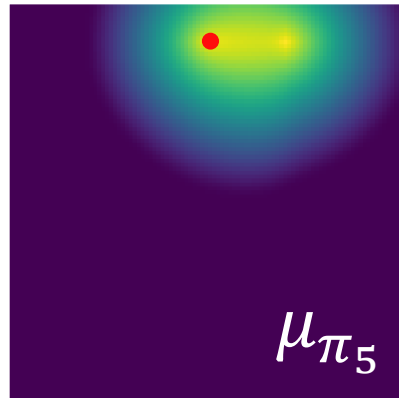
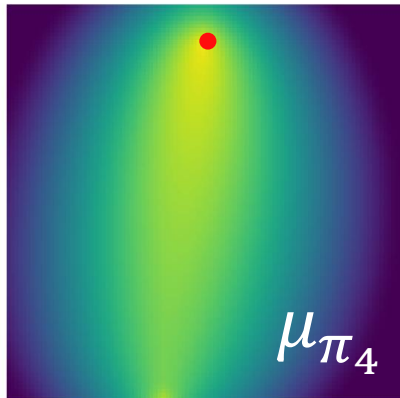
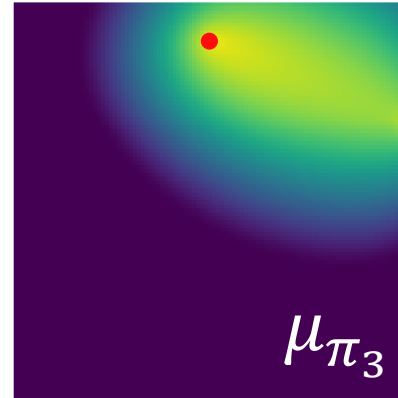
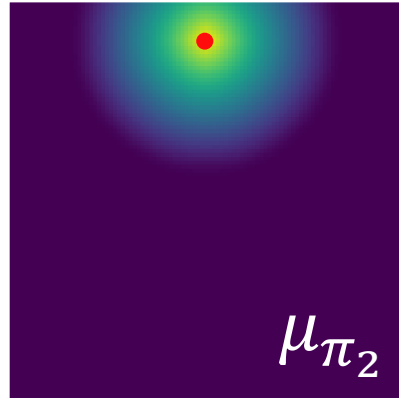
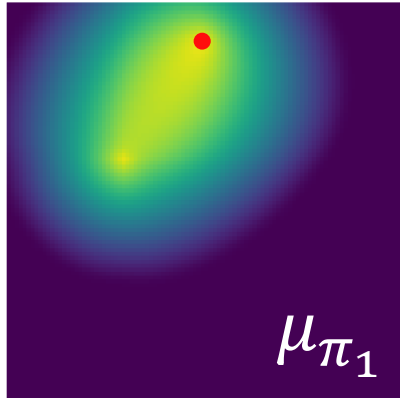
Observe: the discounted return of policy π is

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Discounted return is linear in μ_π :

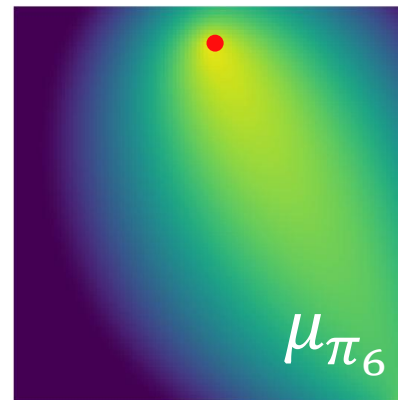
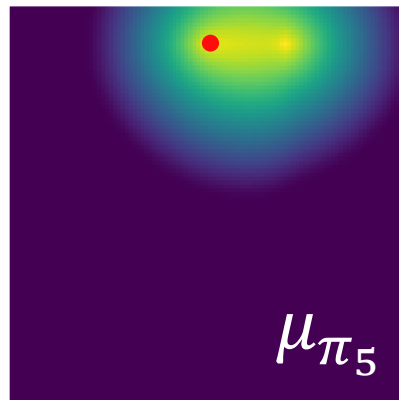
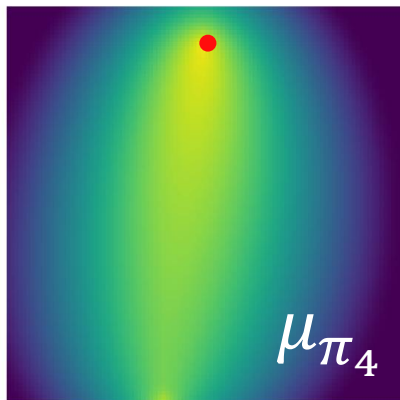
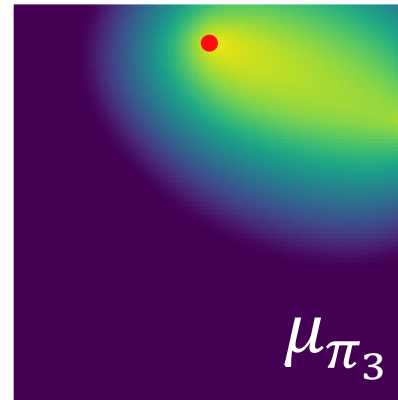
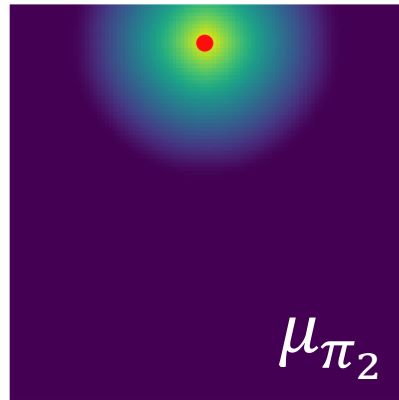
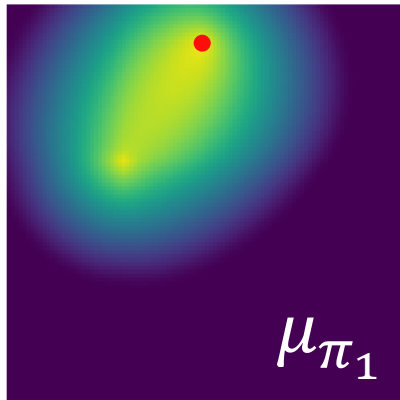
$$(1-\gamma)\rho^\pi = \langle \mu_\pi, r \rangle \stackrel{\text{def}}{=} \sum_{x,a} \mu_\pi(x, a) r(x, a)$$

EXAMPLE: 2D STATE SPACE



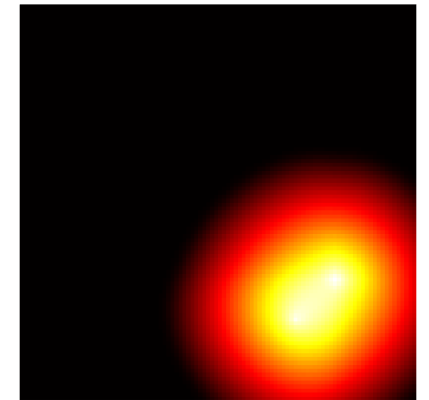
• = initial state

EXAMPLE: 2D STATE SPACE



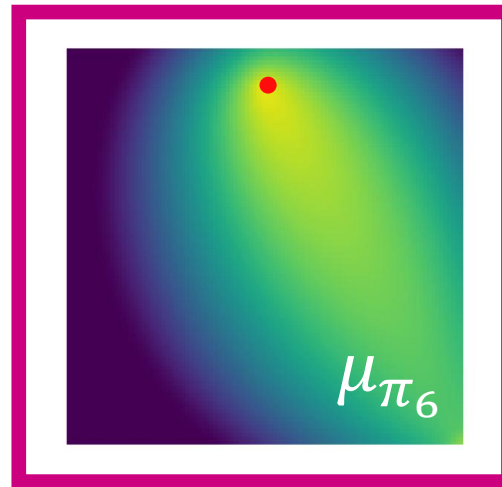
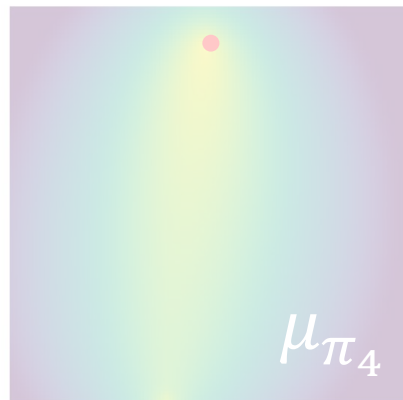
$\times$

reward function



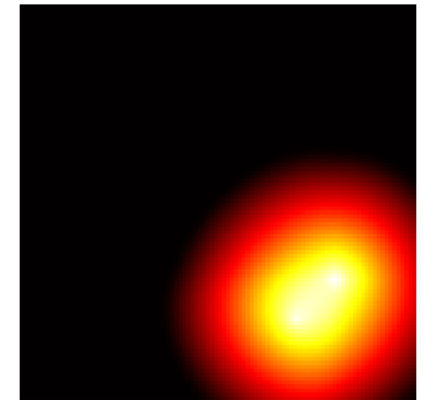
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EXAMPLE: 2D STATE SPACE

How can we do this efficiently over the set of all policies?

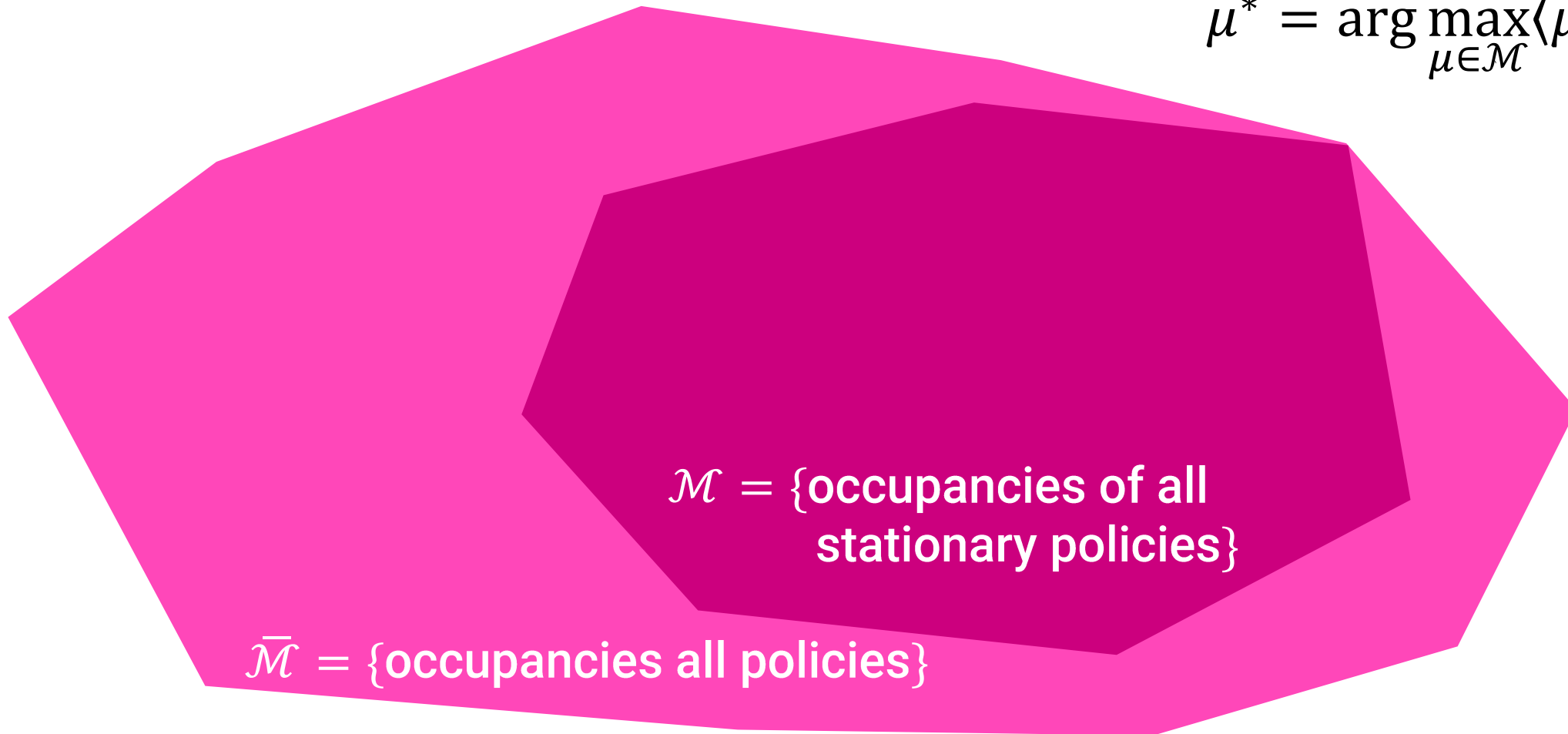


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OPTIMIZATION OVER THE OCCUPANCY SPACE

We would like to compute:

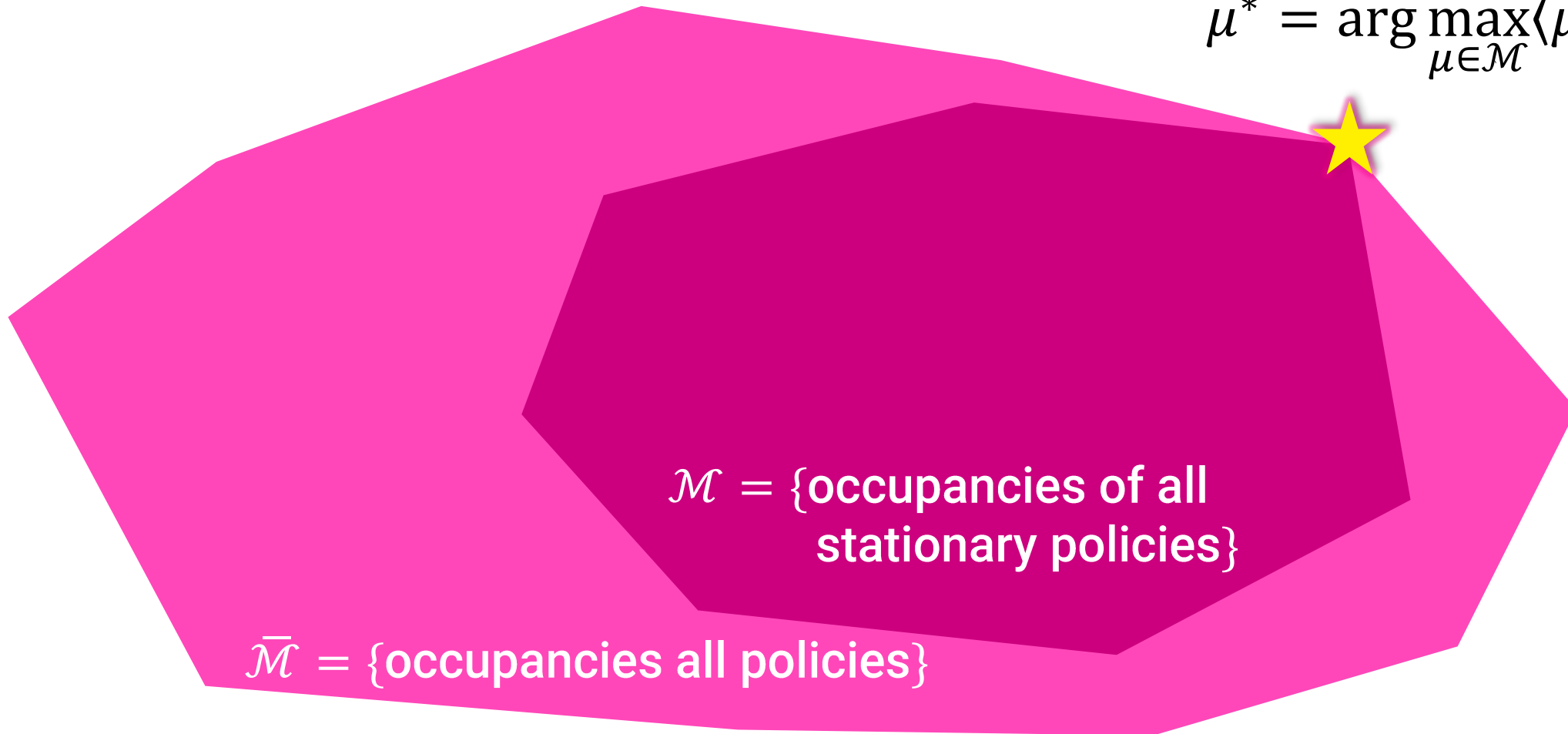
$$\mu^* = \arg \max_{\mu \in \mathcal{M}} \langle \mu, r \rangle$$



OPTIMIZATION OVER THE OCCUPANCY SPACE

We would like to compute:

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USEFUL NOTATION, PT. 1

- We will represent
 - state-action occupancy measures as vectors in $\mathbb{R}^{\mathcal{X} \times \mathcal{A}}$
 - state occupancy measures as vectors in $\mathbb{R}^{\mathcal{X}}$
 - action-value functions as vectors in $\mathbb{R}^{\mathcal{X} \times \mathcal{A}}$
 - value functions as vectors in $\mathbb{R}^{\mathcal{X}}$
- Expectations are written as scalar products:
$$\langle \mu, f \rangle \stackrel{\text{def}}{=} \mathbb{E}_{X, A \sim \mu}[f(X, A)]$$
$$\langle \nu, g \rangle \stackrel{\text{def}}{=} \mathbb{E}_{X \sim \nu}[g(X)]$$

USEFUL NOTATION, PT. 2

- MDP operators:

- Conditional expectation operator $P: \mathbb{R}^{\mathcal{X}} \rightarrow \mathbb{R}^{\mathcal{X} \times \mathcal{A}}$

$$(PV)(x, a) = \sum_{x'} P(x'|x, a)V(x')$$

- Transition operator $P^\top: \mathbb{R}^{\mathcal{X} \times \mathcal{A}} \rightarrow \mathbb{R}^{\mathcal{X}}$

$$(P^\top \mu)(x) = \sum_{x', a'} \mu(x', a')P(x|x', a')$$

- Broadcasting operator $E: \mathbb{R}^{\mathcal{X}} \rightarrow \mathbb{R}^{\mathcal{X} \times \mathcal{A}}$

$$(EV)(x, a) = V(x)$$

- Marginalization operator $E^\top: \mathbb{R}^{\mathcal{X} \times \mathcal{A}} \rightarrow \mathbb{R}^{\mathcal{X}}$

$$(E^\top \mu)(x) = \sum_a \mu(x, a)$$

USEFUL NOTATION, PT. 3

- Operators for a fixed stochastic policy $\pi: \mathcal{X} \rightarrow \Delta_{\mathcal{A}}$:

- Conditional expectation operator $P_{\pi}: \mathbb{R}^{\mathcal{X}} \rightarrow \mathbb{R}^{\mathcal{X}}$

$$(P_{\pi}V)(x) = \sum_{x'} \sum_a \pi(a|x)P(x'|x, a)V(x')$$

- Transition operator $P_{\pi}^{\top}: \mathbb{R}^{\mathcal{X}} \rightarrow \mathbb{R}^{\mathcal{X}}$

$$(P_{\pi}^{\top}v)(x) = \sum_{x', a'} v(x')\pi(a'|x')P(x|x', a')$$

STATE OCCUPANCIES

Definition 1

The state occupancy measure ν_π of policy π is the distribution

$$\nu_\pi(x) = (1 - \gamma) \sum_{t=0}^{\infty} \mathbb{P}_\pi[X_t = x]$$

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Theorem 1

$$v_\pi = \gamma P_\pi^\top v_\pi + (1 - \gamma)v_0$$

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Proof. Let $\nu_t(x) = \mathbb{P}_\pi[X_t = x]$ and notice that $\nu_t = P_\pi^\top \nu_{t-1}$. Then,
$$\nu_\pi = (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t \nu_t = (1 - \gamma) \sum_{t=0}^{\infty} (\gamma P_\pi^\top)^t \nu_0$$

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FLOW EQUATIONS

Theorem 2

ν is the state-occupancy measure of π **if and only if** it satisfies

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Proof. The condition can be rewritten as the linear system

$$(\text{Id} - \gamma P_{\pi}^{\top}) \nu = (1 - \gamma) \nu_0,$$

which has a unique solution since $(\text{Id} - \gamma P_{\pi}^{\top})$ is invertible.

(By the Perron–Frobenius theorem.)

STATE-ACTION OCCUPANCIES

Definition 2

The occupancy measure μ_π of policy π is the distribution

$$\mu_\pi(x, a) = (1 - \gamma) \sum_{t=0}^{\infty} \mathbb{P}_\pi[X_t = x, A_t = a]$$

STATE-ACTION OCCUPANCIES

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Theorem 3

$$E^\top \mu_\pi = \gamma P^\top \mu_\pi + (1 - \gamma) \nu_0$$

PROOF

Theorem 3

$$E^\top \mu_\pi = \gamma P^\top \mu_\pi + (1 - \gamma) v_0$$

Proof.

$$\begin{aligned} \sum_a \mu_\pi(x, a) &= (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t \mathbb{P}_\pi[X_t = x] \\ &= (1 - \gamma) v_0(x) + \sum_{t=1}^{\infty} \sum_{x', a'} \gamma^t \mathbb{P}_\pi[X_t = x, (X_{t-1}, A_{t-1}) = (x', a')] \\ &= (1 - \gamma) v_0(x) + \sum_{t=1}^{\infty} \sum_{x', a'} \gamma^t P(x|x', a') \mathbb{P}_\pi[(X_{t-1}, A_{t-1}) = (x', a')] \\ &= (1 - \gamma) v_0(x) + \sum_{x', a'} P(x|x', a') \sum_{t=1}^{\infty} \gamma^t \mathbb{P}_\pi[(X_{t-1}, A_{t-1}) = (x', a')] \\ &= (1 - \gamma) v_0(x) + \sum_{x', a'} P(x|x', a') \mu_\pi(x', a') \end{aligned}$$

PROOF

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SUPER IMPORTANT:

this claim & its proof holds for all policies, not just stationary ones!!!

FLOW EQUATIONS

Theorem 4

μ is a valid occupancy measure **if and only if** it satisfies

$$E^\top \mu = \gamma P^\top \mu + (1 - \gamma) v_0$$

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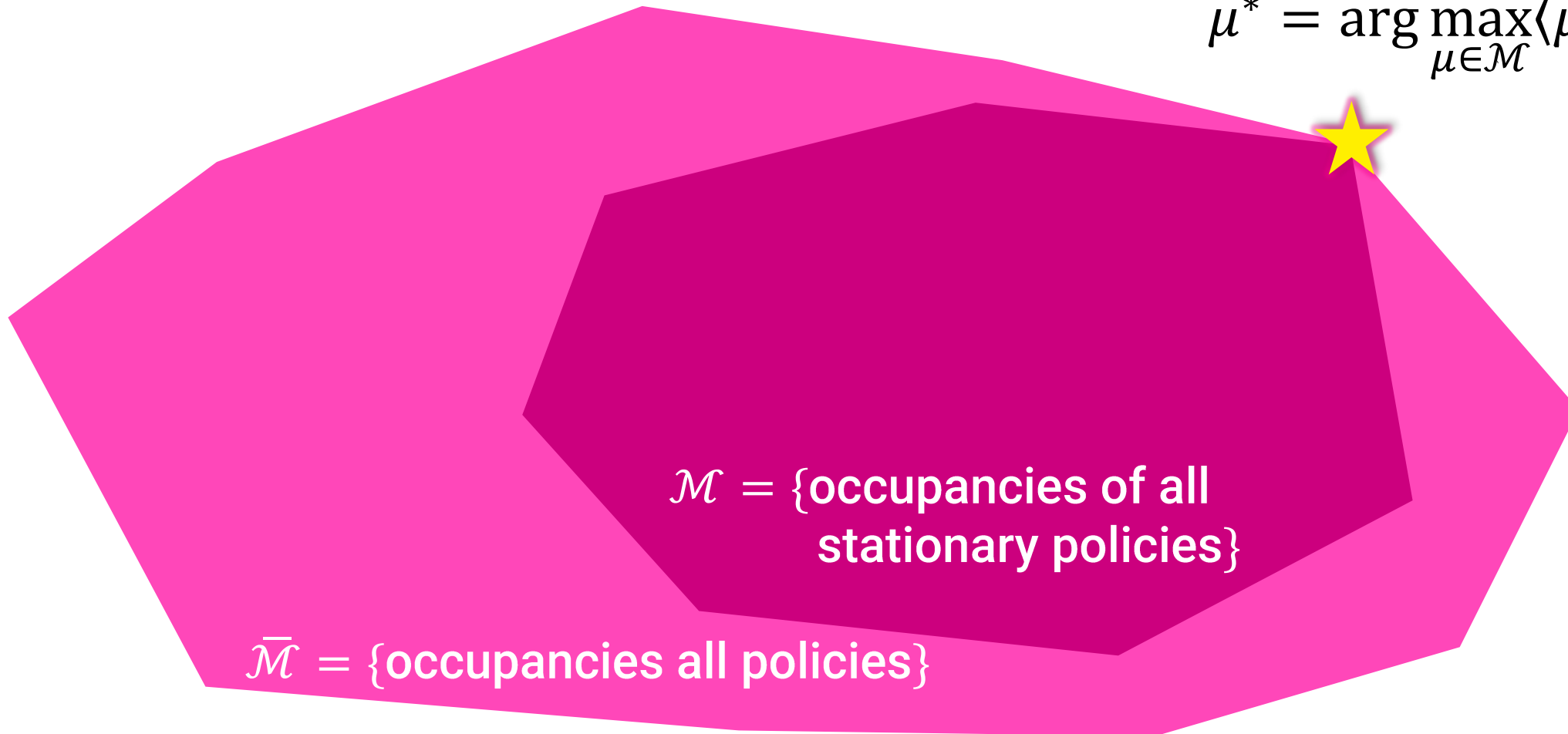
Proof. We already have “ $\Rightarrow$ ”, so we only need “ $\Leftarrow$ ”:

- Let μ satisfy the flow condition stated in the theorem.
- Let $v = E^\top \mu$ and $\pi(a|x) = \frac{\mu(x,a)}{v(x)}$.
- Observe that $P^\top \mu = P_\pi^\top v$, so the condition is equivalent to $v = \gamma P_\pi^\top v + (1 - \gamma)v_0$. This has v_π as its unique soln. (Thm 2)
- $\mu(x, a) = v_\pi(x)\pi(a|x) = \sum_{t=0}^{\infty} \gamma^t \mathbb{P}_\pi[X_t = x, A_t = a] = \mu_\pi(x, a)$

AN UNEXPECTED WIN

We would like to compute:

$$\mu^* = \arg \max_{\mu \in \mathcal{M}} \langle \mu, r \rangle$$



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$\mathcal{M} = \{\text{occupancies of all stationary policies}\}$

$\bar{\mathcal{M}} = \{\text{occupancies all policies}\}$

AN UNEXPECTED WIN

We would like to compute:

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Formal proof that stationary policies are sufficient for maximizing expected reward!

$\mathcal{M} = \{\text{occupancies of all stationary policies}\}$

$\bar{\mathcal{M}} = \{\text{occupancies all policies}\}$

THE LP FORMULATION

Linear Programming for MDPs

$$\begin{array}{ll}\text{maximize} & \langle \mu, r \rangle \\ \text{subject to} & E^\top \mu = \gamma P^\top \mu + (1 - \gamma)v_0\end{array}$$

THE LP FORMULATION

Linear Programming for MDPs

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- Optimal policy π^* can be extracted from solution μ^* as

$$\pi^*(a|x) = \frac{\mu^*(x, a)}{\sum_a \mu^*(x, a')}$$

THE LP FORMULATION

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Dual Linear Program for MDPs

$$\begin{array}{ll}\text{minimize} & (1 - \gamma)\langle v_0, V \rangle \\ \text{subject to} & EV \geq r + \gamma PV\end{array}$$

HOW DO WE DERIVE ALGORITHMS FROM THIS?

Spoiler: by doing something different 😊
(For actual answer, see my RLSS 2023 lecture)

THE PERFORMANCE DIFFERENCE LEMMA

THE STAR OF THE SHOW

Performance Difference Lemma

(Howard, 1960, Langford & Kakade, 2006)

Let π and π' be two stationary policies. Then,

$$\rho^\pi - \rho^{\pi'} = \frac{1}{1 - \gamma} \langle \mu_\pi, Q_{\pi'} - EV_{\pi'} \rangle$$

THE STAR OF THE SHOW

Performance Difference Lemma

(Howard, 1960, Langford & Kakade, 2006)

Let π and π' be two stationary policies. Then,

$$\rho^\pi - \rho^{\pi'} = \frac{1}{1 - \gamma} \langle \mu_\pi, Q_{\pi'} - EV_{\pi'} \rangle$$

Proof.

- Bellman equations for π' : $Q_{\pi'} = r + \gamma P V_{\pi'}$
- Multiply both sides with μ_π and reorder:

$$\langle \mu_\pi, r \rangle = \langle \mu_\pi, Q_{\pi'} - \gamma P V_{\pi'} \rangle$$

- Flow equations for π :

$$\langle \mu_\pi, \gamma P V_{\pi'} \rangle = \langle \gamma P^\top \mu_\pi, V_{\pi'} \rangle = \langle E^\top \mu_\pi - (1 - \gamma)v_0, V_{\pi'} \rangle$$

- Notice that $\rho^\pi = \frac{\langle \mu_\pi, r \rangle}{1 - \gamma}$ and $\rho^{\pi'} = \langle v_0, V_{\pi'} \rangle$



A GLORIOUS TOOL OF MDP THEORY

- Combines both sides of DP theory: Bellman & flow equations
- Useful for proving fundamental results:
 - Policy improvement theorem
 - Policy gradient theorem
- Useful for developing and analyzing advanced RL methods:
 - Soft policy iteration / natural policy gradient
 - Optimistic exploration

APPLICATIONS

- Policy improvement theorem
- Policy gradient theorem
- Soft policy improvement

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A CLASSIC RESULT

Policy Improvement Theorem

(Howard, 1960)

Let π be a policy and π' be greedy wrt. Q^π . Then,
either $\rho^{\pi'} > \rho^\pi$ holds or π is optimal.

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Proof.

- PD Lemma: $\rho^{\pi'} - \rho^\pi = \frac{1}{1-\gamma} \langle \mu_{\pi'}, Q_\pi - EV_\pi \rangle$
- Since π' is greedy, we have $\sum_a \pi'(a|x) Q_\pi(x, a) \geq V_\pi(x)$
- If there is one state where ineq. is strict, we have $\rho^{\pi'} > \rho^\pi$
- Otherwise, we have $\max_a Q_\pi(x, a) = V_\pi(x)$ for all x and thus $V_\pi = V^*$

APPLICATIONS

- Policy improvement theorem
- Policy gradient theorem
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ANOTHER BANGER

Policy Gradient Theorem

Let π_θ be a parametric policy with $\pi_\theta(a|x) > 0$ continuously differentiable wrt. $\theta \in \mathbb{R}^d$ for all x, a .

Then, the gradient of $\rho_\theta = \rho_{\pi_\theta}$ satisfies

$$\nabla_\theta \rho_\theta = \mathbb{E}_{X, A \sim \mu_{\pi_\theta}} [\nabla_\theta \log \pi_\theta(A|X) Q_{\pi_\theta}(X, A)]$$

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Proof.

- Let us fix $i \in \{1, 2, \dots, d\}$ and let $e_i \in \mathbb{R}^d$ be the corresponding unit vector.
- Let $\pi = \pi_\theta$ and $\pi'_\varepsilon = \pi_{\theta + \varepsilon e_i}$ for some $\varepsilon \in \mathbb{R}$
- The i th component of the gradient we are looking for is:

$$\partial \rho_\theta / \partial \theta_i = \lim_{\varepsilon \rightarrow 0} \frac{\rho_\theta - \rho_{\theta + \varepsilon e_i}}{\varepsilon}$$

PROOF CONT'D

- Unleash the PDL:

$$\begin{aligned}\frac{\rho_{\theta} - \rho_{\theta + \varepsilon e_i}}{\varepsilon} &= \frac{1}{\varepsilon} \langle \mu_{\pi_{\theta}}, Q_{\pi'_{\varepsilon}} - EV_{\pi'_{\varepsilon}} \rangle \\ &= \sum_x \nu_{\pi_{\theta}}(x) \sum_a \frac{(\pi_{\theta}(a|x) - \pi'_{\varepsilon}(a|x))}{\varepsilon} Q_{\pi'_{\varepsilon}}(x, a)\end{aligned}$$

PROOF CONT'D

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APPLICATIONS

- Policy improvement theorem
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SOFT POLICY IMPROVEMENT

- Policy gradient methods work with the wrong geometry:

$$\begin{aligned}\theta_{k+1} &= \theta_k + \alpha \nabla_{\theta} \rho_{\pi_{\theta_k}} \\ &\approx \arg \min_{\theta} \rho_{\pi_{\theta}} + \|\theta - \theta_k\|_2^2\end{aligned}$$

- “Linearize wrt. θ and optimize locally”
- A better idea: “linearize wrt. π and optimize locally”

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- A better idea: “linearize wrt. π and optimize locally”

Linearized PD Lemma

The first-order Taylor approximation of $\rho_{\pi} - \rho_{\pi_k}$ wrt. π around π_k is written as

$$\tilde{\mathcal{L}}_k(\pi) = \mathbb{E}_{X \sim \nu_{\pi_k}} \left[\sum_a \pi(a|x) \left(Q_{\pi_k}(X, a) - V_{\pi_k}(X) \right) \right]$$

PROOF

- Use PG theorem to compute derivatives:

$$\left. \frac{\partial \rho_\pi}{\partial \pi(a|x)} \right|_{\pi_k} = v_{\pi_k}(x) \left(Q_{\pi_k}(x, a) - V_{\pi_k}(x) \right)$$

- Plug this into Taylor expansion formula:

$$\begin{aligned} \rho_\pi &\approx \rho_{\pi_k} + \sum_{x,a} \left(\pi(a|x) - \pi_k(a|x) \right) \cdot \left. \frac{\partial \rho_\pi}{\partial \pi(a|x)} \right|_{\pi_k} \\ &= \rho_{\pi_k} + \sum_{x,a} \left(\pi(a|x) - \pi_k(a|x) \right) v_{\pi_k}(x) \left(Q_{\pi_k}(x, a) - V_{\pi_k}(x) \right) \end{aligned}$$

- Notice $\sum_a \pi_k(a|x) Q_{\pi_k}(x, a) = V_{\pi_k}(x)$

ALGORITHM IDEA

- Consider optimization landscape



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own ugly figures

ALGORITHM IDEA

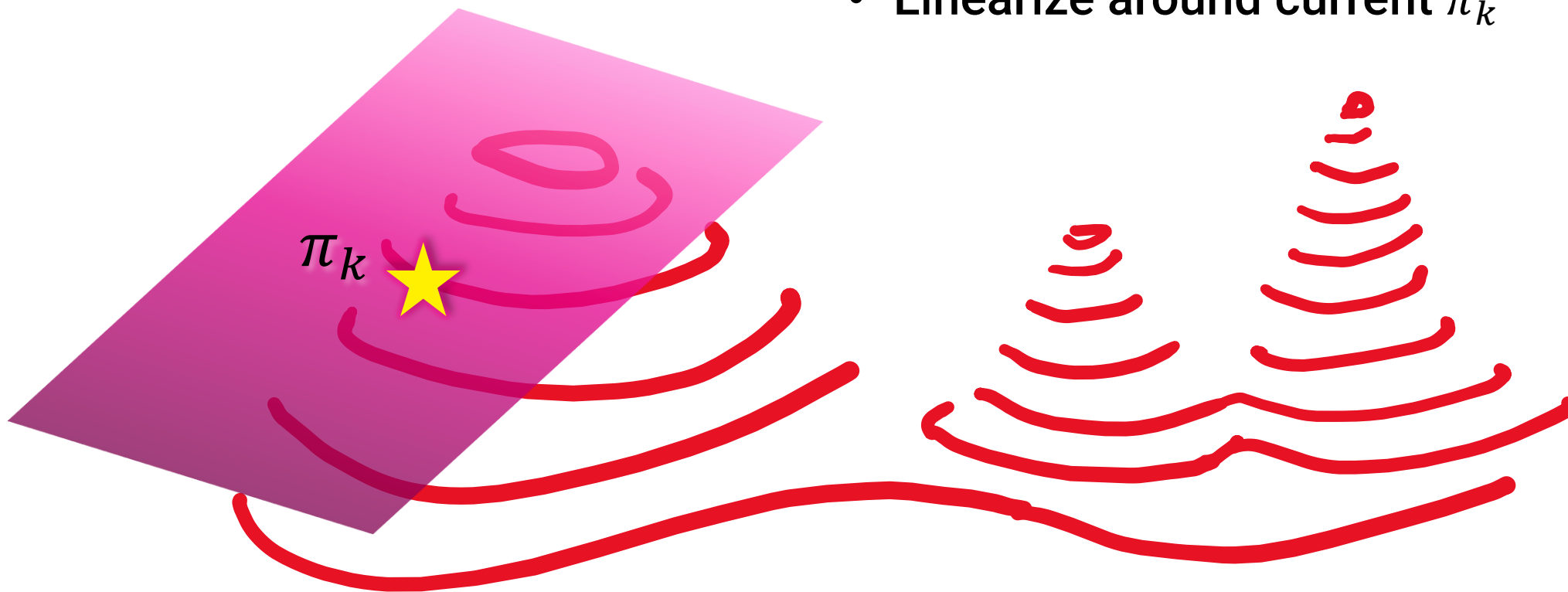
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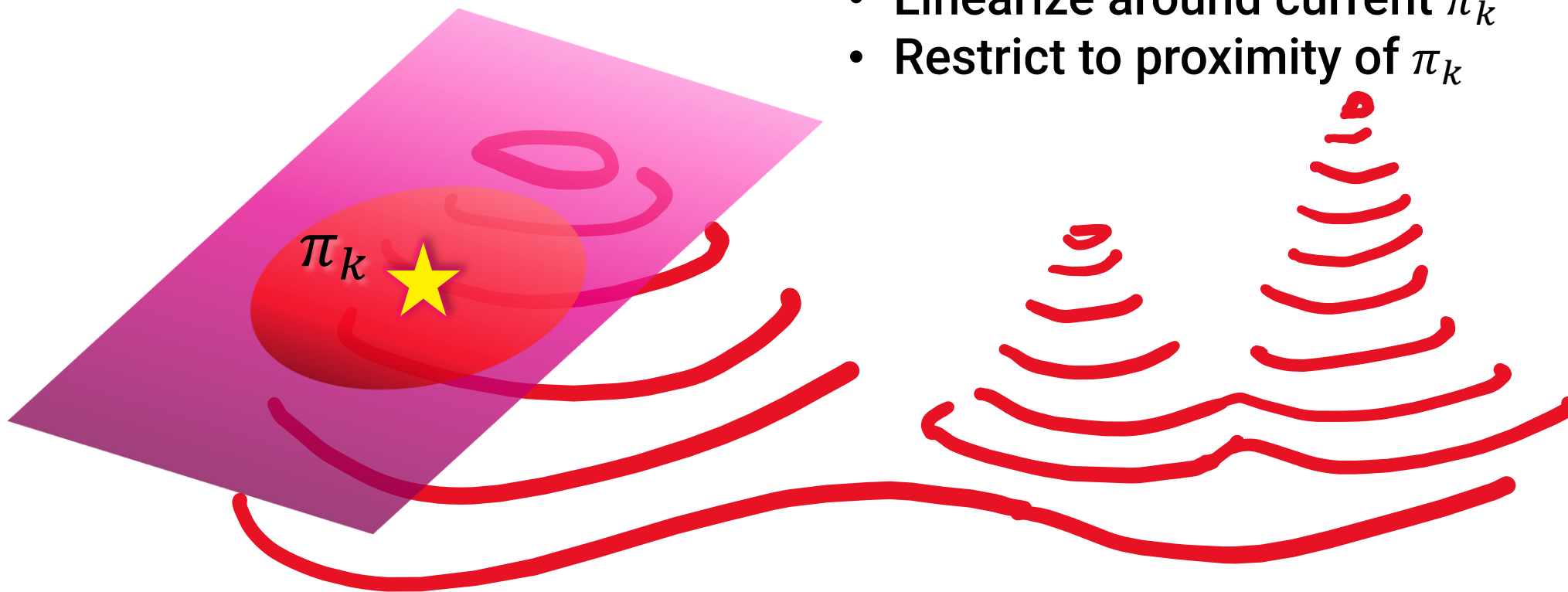
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ALGORITHM IDEA

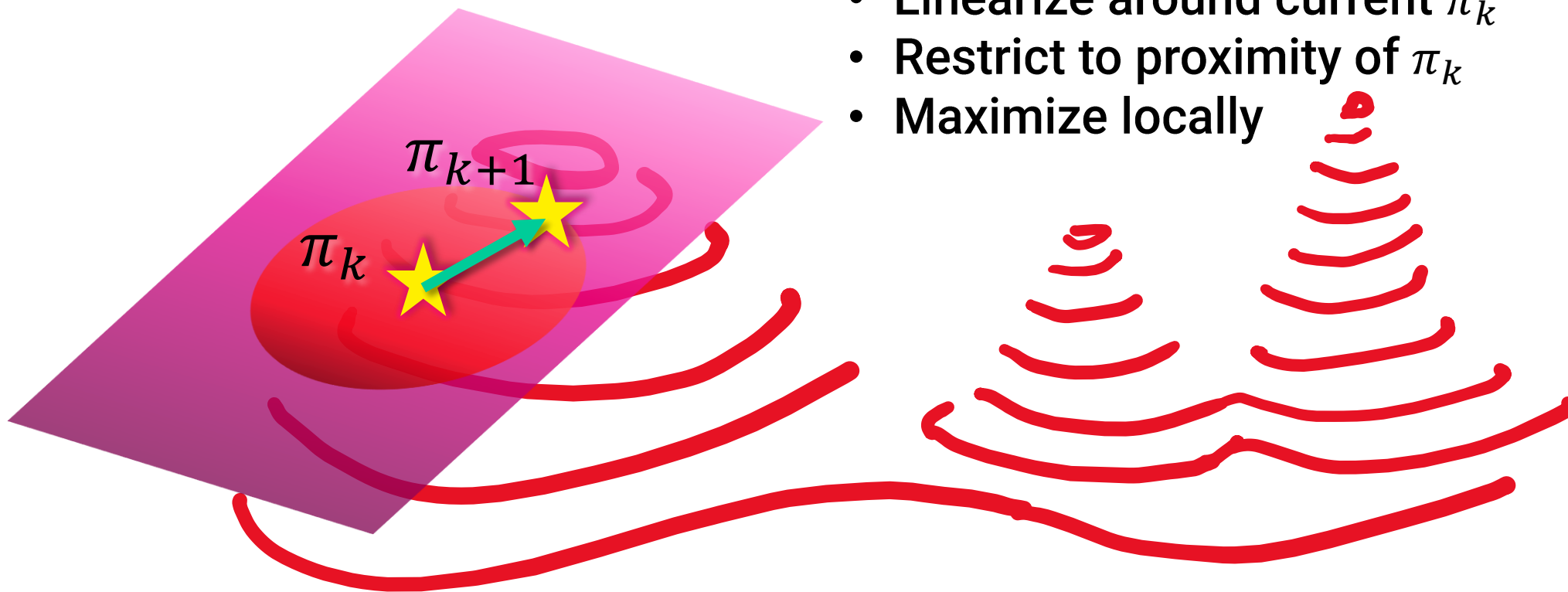
- Consider optimization landscape
- Linearize around current π_k
- Restrict to proximity of π_k



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ALGORITHM IDEA

- Consider optimization landscape
- Linearize around current π_k
- Restrict to proximity of π_k
- Maximize locally



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SOME EXAMPLES

- Natural policy gradient / regularized PI / POLITEX:
 - Add entropy regularization before linearizing objective
 - Closed form updates: $\pi_{k+1}(a|x) \propto \pi_k(a|x)e^{\alpha Q_{\pi_k}(x,a)}$
- Trust-region policy optimization (TRPO):
 - Same as above, with some more hacks
 - Importance-weighted Monte Carlo to estimate Q_{π_k}
 - Replace additive regularization by hard constraint
 - Swap the arguments of the relative entropy penalty
- Proximal policy optimization (PPO):
 - Ditch regularization
 - Enforce locality by truncating importance weights

CONCLUSION



OCCUPANCIES & PD LEMMA

- Useful tools for
 - proving fundamental results
 - designing and analyzing RL methods
- More common than you might think! (“find the PD lemma”)
- Often more convenient than standard DP theory (e.g., in constrained / multi-objective / convex RL)
- Basis for more advanced “primal-dual” methods



Thanks!!



APPENDIX

MIRROR DESCENT

Occupancy mirror descent

Init: policy π_0 , occupancy $\mu_0 = \mu_{\pi_0}$

For $k = 0, 1, \dots$ **repeat**

- **Update occupancy:**

$$\mu_{k+1} = \arg \max_{\mu \in \mathcal{M}} \langle \mu, r \rangle - \frac{1}{\alpha} D(\mu | \mu_k)$$

Closed form updates:

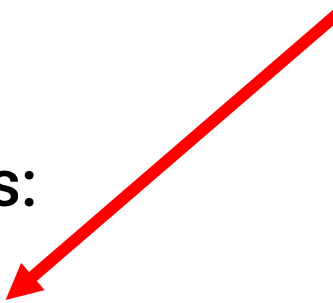
$$\pi_{k+1}(a|x) \propto \pi_k(a|x) e^{\alpha \tilde{Q}_k(x,a)},$$

where $\tilde{Q}_k$ is the solution of the regularized Bellman equations:

$$\tilde{Q}_k = r + \gamma P_{\pi_k} \tilde{V}_k$$

$$\tilde{V}_k = \text{softmax}(\tilde{Q}_k, \pi_k)$$

Painful to compute :/



POLICY MIRROR DESCENT

- Inspired by occupancy MD, we aim to optimize:

$$\mathcal{L}_k(\pi) = \rho_\pi - \frac{1}{\alpha} D(\pi | \pi_k)$$

- For better practicality, linearize $\mathcal{L}_k$ around π_k :

$$\begin{aligned}\tilde{\mathcal{L}}_k(\pi) &= \mathcal{L}_k(\pi_k) + \langle \pi - \pi_k, \nabla_\pi \mathcal{L}_k(\pi_k) \rangle \\ &= \rho_{\pi_k} + \mathbb{E}_{X \sim \nu_{\pi_k}} \left[\sum_a \pi(a|x) \left(Q_{\pi_k}(X, a) - V_{\pi_k}(X) - \frac{1}{\alpha} \log \frac{\pi(a|X)}{\pi_k(a|X)} \right) \right]\end{aligned}$$

- Optimizer has closed form:

$$\pi_{k+1}(a|x) \propto \pi_k(a|x) e^{\alpha Q_{\pi_k}(x,a)}$$

Much less painful to compute!

MIRROR DESCENT

Occupancy mirror descent

Init: policy π_0 , occupancy $\mu_0 = \mu_{\pi_0}$

For $k = 0, 1, \dots$ **repeat**

- Update occupancy:

$$\mu_{k+1} = \arg \max_{\mu \in \mathcal{M}} \langle \mu, r \rangle - \frac{1}{\alpha} D(\mu | \mu_k)$$

Policy mirror descent

Init: policy π_0 , occupancy $\mu_0 = \mu_{\pi_0}$

For $k = 0, 1, \dots$ **repeat**

- Evaluate policy: $Q_k = Q_{\pi_k}$

- Update policy:

$$\pi_{k+1}(a|x) \propto \pi_k(a|x) e^{\alpha Q_{\pi_k}(x,a)}$$

A PRACTICAL ALGORITHM: TRPO

- Notice that the only part of the linearized objective $\tilde{\mathcal{L}}_k(\pi)$ that depends on π is:

$$\hat{\mathcal{L}}_k(\pi) = \mathbb{E}_{X \sim \nu_{\pi_k}} \left[\sum_a \pi(a|x) \left(Q_{\pi_k}(X, a) - \frac{1}{\alpha} \log \frac{\pi(a|X)}{\pi_k(a|X)} \right) \right]$$

- This can be estimated based on on-policy data from π_k !
- **Algorithm idea:** parametrize π_θ and then repeatedly update policy parameters as:

$$\theta_{k+1} = \arg \min_{\theta} \hat{\mathcal{L}}_k(\theta)$$

TRPO = this idea + a bunch of weird unjustified tricks

e.g. swapping the arguments of the KL div, replacing penalty with a hard constraint,...

FROM TRPO TO PPO

- Proximal policy optimization:
 - Remove entropy regularization and hard constraint
 - Importance-weighted estimate of the linearized objective $\hat{\mathcal{L}}_k(\pi)$
 - Locally optimize this loss, with locality enforced by clipping the objective when π is changed too much